

Appendix J: GAN Surrogate Landscapes and Interaction Analysis

For insertion into the BIF paper appendix

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J. GAN Surrogate Landscapes and Interaction Analysis

J.1 Child Hyperparameter Landscapes

Figure 1 shows the FID landscape for each child sub-problem in both surrogate tasks.

Standard task (top row). The generator landscape (a) is smooth and dominated by learning rate: FID degrades sharply for $\text{LR}_{\text{gen}} \geq 5 \times 10^{-3}$, while weight decay has a comparatively mild effect. The discriminator landscape (b) is more complex, with a narrow basin of good configurations at intermediate learning rates (2×10^{-4} – 10^{-3}) and low-to-moderate dropout (0.0–0.3). High learning rates or high dropout both cause catastrophic FID increases.

SpectralNorm task (bottom row). The generator landscape (c) retains the same qualitative structure as in the standard task—low learning rates are preferred—consistent with the shared generator architecture. The discriminator landscape (d), however, shifts noticeably: Spectral Normalization stabilizes training, broadening the region of acceptable configurations and reducing the dropout sensitivity. This landscape shift is precisely the change that the modularity experiment exploits: the generator child transfers directly, while the discriminator child must be re-learned.

J.2 Non-Linear Interaction Between Generator and Discriminator

BIF assumes a quasi-linear additive interaction between children (Section 3): $F(x) \approx f_{\text{gen}}(x_{\text{gen}}) + f_{\text{disc}}(x_{\text{disc}}) + c$. To quantify how well this assumption holds for the GAN surrogate, we perform an ANOVA-style variance decomposition. Let Y_{ij} denote the neg-FID for generator configuration i and discriminator configuration j . The additive reconstruction is $\hat{Y}_{ij} = \bar{Y}_{i.} + \bar{Y}_{.j} - \bar{Y}_{..}$, where $\bar{Y}_{i.}$, $\bar{Y}_{.j}$, and $\bar{Y}_{..}$ are the row, column, and grand means respectively. The residual $R_{ij} = Y_{ij} - \hat{Y}_{ij}$ captures the non-additive (interaction) component.

Table 1 summarizes the decomposition. For the **standard task**, the additive model explains only 45.5% of total variance ($R^2 = 0.455$), with 54.5% attributable to gen-disc interaction. This substantial non-linearity means that the optimal generator configuration depends on which discriminator is used, and vice versa — a challenging scenario for any additive-prior method. For the **SpectralNorm task**, additivity improves to $R^2 = 0.804$, with only 19.6% interaction variance, likely because Spectral Normalization reduces the discriminator’s sensitivity to generator choices.

Figure 2 visualizes this decomposition. The residual heatmaps (c, f) reveal structured interaction patterns: in the standard task, specific gen-disc combinations produce FID values far from what either marginal would predict, particularly at high generator learning rates. Despite this violation of additivity, BIF’s performance remains competitive (Table 1), demonstrating the robustness noted in Appendix G.

Task	Additive R^2	Interaction (% of var)
Standard (BatchNorm)	0.455	54.5%
SpectralNorm	0.804	19.6%

Table 1: Variance decomposition of the GAN surrogates. The additive model $\hat{Y}_{ij} = \bar{Y}_i + \bar{Y}_j - \bar{Y}$ is compared to the true neg-FID table. The standard task exhibits strong non-linear gen-disc interaction (54.5% of variance), while SpectralNorm reduces this to 19.6%.

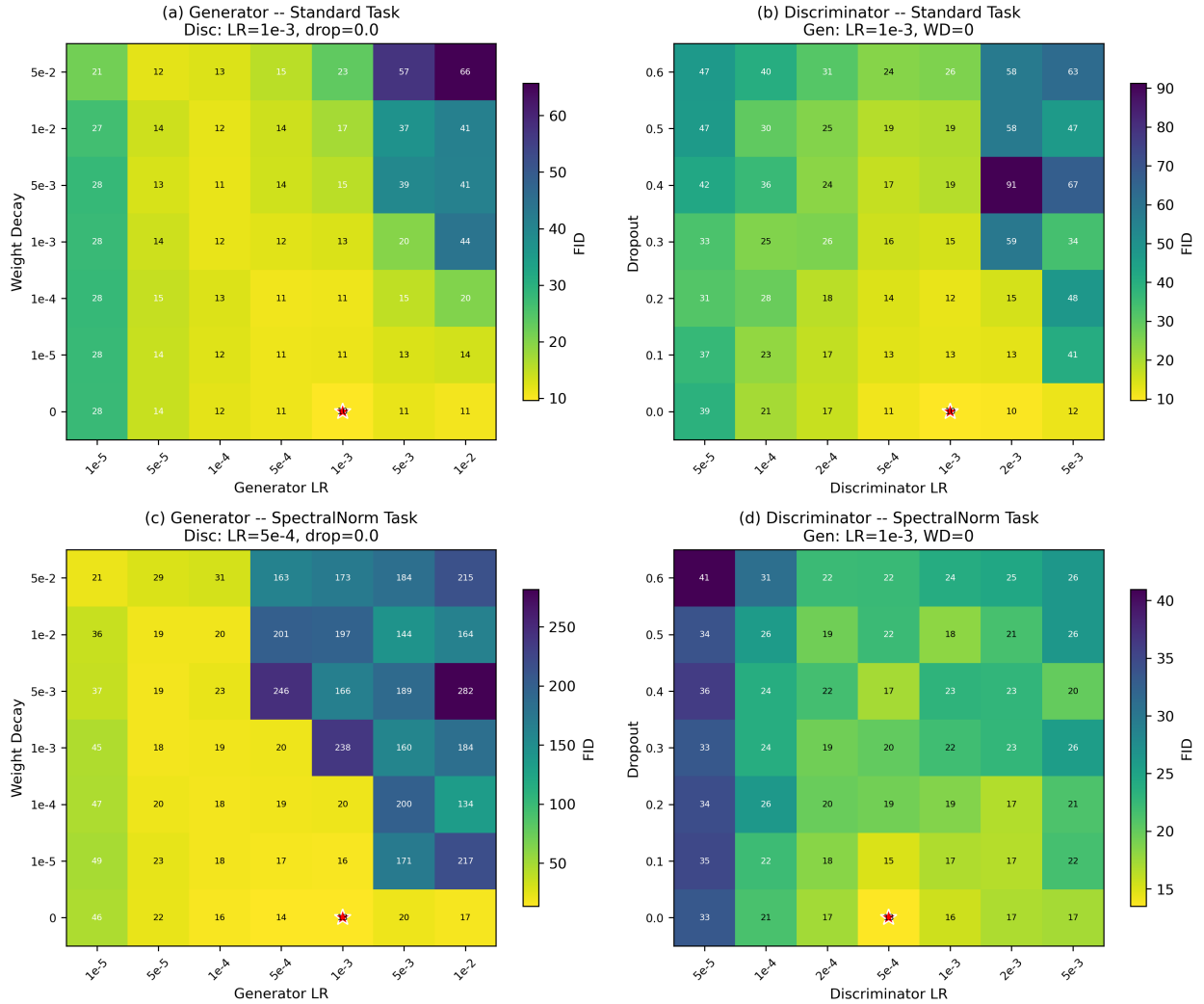


Figure 1: Child hyperparameter landscapes for the GAN surrogate. **Top:** Standard task (BatchNorm discriminator). **Bottom:** SpectralNorm task. Each heatmap shows FID (lower = better) as a function of the child’s two hyperparameters, with the other child fixed at its task-specific optimum. Red stars mark the global optimum.

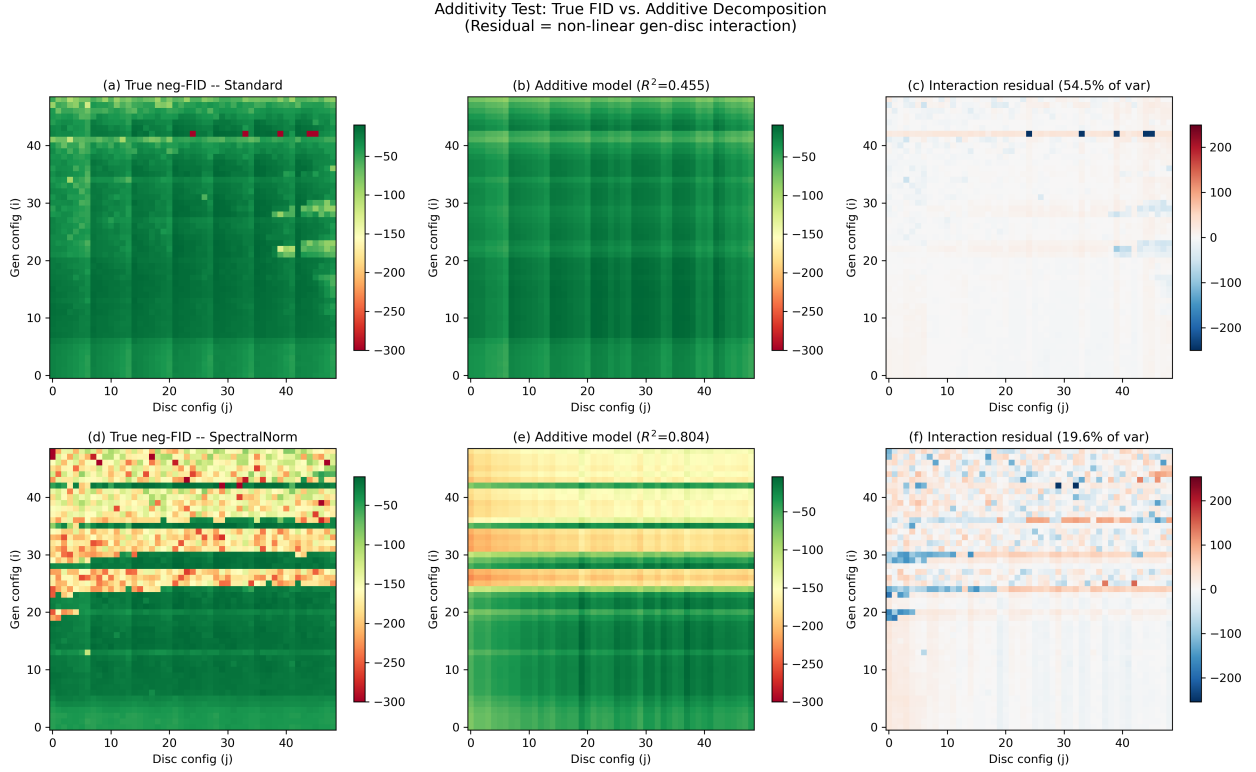


Figure 2: Additivity test for the GAN surrogates. **Left:** True neg-FID landscape (49 gen configs $\times$ 49 disc configs). **Center:** Additive reconstruction from marginal means. **Right:** Residual (interaction component), plotted on a diverging colormap (red/blue = positive/negative deviation from additivity). **Top row:** Standard task ($R^2 = 0.455$, 54.5% interaction). **Bottom row:** SpectralNorm task ($R^2 = 0.804$, 19.6% interaction).